

Examination and Academic Regulations for the Master's Program Management at the Heilbronn Campus at the Technische Universität München

**dated 15. November 2021
as amended in the third amendment from 16th August 2023**

In accordance with Art. 13 (1) sentence 2 in conjunction with Art. 58 (1) sentence 1, Art. 61 (2) sentence 1 and Art. 43 (5) of the Bavarian Higher Education Act (BayHSchG), the Technical University of Munich issues the following statutes:

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§ 34

Applicability, Academic Titles and related degrees

- (1) ¹The subject examination and study regulations for the Master's degree program in Management at the Heilbronn Campus at the Technical University of Munich (FPSO) s the complement the General Examination and Study Regulations for Bachelor's and Master's Degree Programs at the Technical University of Munich (APSO) dated March 18, 2011, as amended. ²The APSO shall take precedence.
- (2) ¹Upon successful completion of the Master's examination the degree „Master of Science“ („M.Sc.“) is awarded. ²The academic title may also be used with the name of the university „(TUM)“.
- (3) ¹The master degree „management“, „industrial engineering“ and „economics for natural sciences“ at the Technische Universität München are related degree programs. ²Where a student transfers from another university to the Technische Universität München, the examination board in charge will decide on the basis of the examination/general academic regulations of that other university if the programs are related.

§ 35

Commencement of Studies, Standard Duration of Studies, ECTS

- (1) The Master's program in Management at the Heilbronn campus of the Technische Universität München generally begins in the winter semester.
- (2) ¹ The number of credits required to obtain the Master's degree in the compulsory and elective areas is 90 (60 semester hours per week), spread over three semesters ²Additionally, 30 credits (maximum six months) have to be achieved for the completion of the Master's thesis pursuant to § 46. The number of examinations in the required and elective subjects to be completed in the Master in Management at Campus Heilbronn of the Technische Universität München according to Appendix 1 (II) is a minimum of 120 credits. ⁴The standard duration of study for the Master's program will be a total of four semesters.

§ 36

Eligibility Requirements

- (1) Eligibility for the master's program in Management at the Heilbronn Campus at the Technical University of Munich is demonstrated by
 1. a qualified Bachelor of Science (B.Sc.) degree obtained after a program of at least six semesters from a domestic or foreign institution of higher education or at least an equivalent degree in engineering or natural sciences, as well as bachelor's degrees the subject area of architecture or comparable programs.
 2. an adequate knowledge of the English language; students whose native language or language of instruction is not English must demonstrate proficiency through an acknowledged language test such as "Test of English as a Foreign Language" (TOEFL) (minimum of 88 points), "International English Language Testing System" (IELTS) (minimum of 6.5 points), or "Cambridge Main Suite of English Examinations"; in the undergraduate program, minimum 12 credits were obtained for examinations administered in English language examination

modules, or a thesis was written in English (minimum of 12 Credits), adequate proficiency in English is deemed proven. Additionally a GMAT-Score of at least 600 also proves adequate proficiency in English.

3. proof of specialist knowledge in the form of a Graduate Management Admission Test (GMAT) with at least 640 points is mandatory for applicants who have completed their first degree in the following countries: China, Bangladesh, India, Egypt, Pakistan; for other applicants with a first degree that was not completed in a signatory state of the Convention on the Recognition of Qualifications concerning Higher Education in the European Region of April 11, 1997 (hereinafter: Lisbon Convention), the submission of the test according to sentence 1 is recommended.
3. passing of the Aptitude Test pursuant to Appendix 2.
- (2) A qualified university degree within the meaning of para. 1 no. 1 exists if there are no significant differences with regard to the competencies (learning outcomes) acquired in the TUM Bachelor's degree programs mentioned in para. 1 no. 1 or comparable degrees and these correspond to the subject-specific requirements of the master's degree program.
- (3) The module catalog of the respective undergraduate degree program is used for the determination according to paragraph 2.

§ 37

Modular Structure, Module Examination, Courses Fields of Study, Language of instruction

- (1) ¹General provisions concerning modules and courses are set forth in §§ 6 und 8 APSO. ²For any changes to the stipulated module provision § 12 (8) of the APSO shall apply.
- (2) ¹The curriculum with the modules in the compulsory and elective areas is listed in Annex 1 (III.).
- (3) ¹The language of instruction in the Master's program in Management at Heilbronn campus of the Technischen Universität München is generally English. ²If students have not provided proof of German language skills at the time of application, the admission will include the requirement that at least one module in which German language skills are acquired on an integrative basis must be successfully completed by the end of the second semester. ³The Examination Board will announce the courses offered in the usual manner. ⁴Voluntary extracurricular courses such as German courses offered by the TUM Language Center are also recognized.

§ 37 a

Advanced Project Study

- (1) ¹In the Master's degree course in Management at the Heilbronn campus of Technische Universität München, students can complete an Advanced Project Studies course. ²The project study entails a project work, which requires an active participation in a practical or research project related to the content of the degree. ³It entails 12 credits and 360 working hours. ⁴The project study will be finalized with a written report and a presentation. ⁵It should be completed by a group consisting of at least two students. ⁶It is supposed to be completed by the end of the third semester. ⁷Through this module it shall be proven that tasks can be solved as a team. ⁸The contribution to be assessed as an examination performance must be clearly recognizable and assessable individually. ⁹This also applies to the individual contribution to the group result. ¹⁰The contribution to be assessed as an examination performance must be clearly recognizable

and assessable on an individual basis. ¹¹This also applies to the individual contribution to the group result. ¹²Section 17 APSO applies to the assessment.

- (2) ¹The Advanced Project Study module has to be supervised by a professor of the TUM department of management. ²Furthermore academic employees can also be appointed as examiners if the respectively applicable requirements of the “Hochschulprüfungsverordnung” are met.

§ 38

Examination Deadlines, Progress Monitoring, Failure to Meet Deadlines

- (1) Examination deadlines, progress monitoring, and failure to meet deadlines are governed by § 10 APSO.
- (2) ¹At least one of the basic module examinations listed in Appendix 1 (II) must be successfully completed by the end of the second semester. In the event of failure to meet deadlines §10 (5) of the APSO shall apply.

§ 39

Examination Board

Pursuant to § 29 of the APSO the board responsible for all decisions concerning examination matters shall be the Master's Examination Board of the School of Management, Technische Universität München.

§ 40

Recognition of Periods of Study, Coursework, and Examination Results

The recognition of periods of study, coursework and examination results is governed by the provision of § 16 of the APSO.

§ 41

Continuous Assessment Procedure/ Examination procedure during the course of study, forms of examination

- (1) ¹In addition to written examinations (Klausuren) and oral examinations, types of assessment pursuant to §§12 und 13 APSO may include exercises, reports, projects, presentations, learning portfolios and/or research papers. ²The specific components of the respective module examination and the competencies to be tested are listed in the module description.³The examination can be carried out as an individual or group examination if the topic is suitable; Section §18 (2) sentences 2 and 3 APSO apply accordingly.
- a) ¹A **Klausur** is a supervised written examination. In these written examinations, students are expected to demonstrate, within a limited amount of time and using predefined methods and resources, their ability to identify problems, find solution strategies and, if required, implement them. ²The duration of Klausuren is provided for in § 12 (7) of the APSO.
- b) ¹**Exercises** are administered to assess a student's ability to complete assigned tasks (e.g. mathematical problems, programming tasks, modeling, designs, etc.) with the aim of applying theoretical content to solve application-related problems. ² Exercises are designed to assess a student's factual and detailed knowledge and its application. ³Practical exercises may be administered in writing, orally, or electronically. ⁴Possible models are assignments,

exercise sheets, programming exercises, (e-)tests, design tasks, posters, tasks as part of university internships, tests, etc.

- c) ¹A **report** is a written record and summary of a learning process for the purpose of presenting the acquired knowledge in a structured way and analyzing the results in the context of a module. ²Students are expected to demonstrate that they have understood all essential aspects and are able to present them in writing. ³Reports may include excursion reports, internship reports, work reports, etc. ⁴The written report may be complemented by a presentation for the purpose of assessing the student's communication competency in presenting scholarly work to an audience.
- d) ¹**Project work** is designed to reach, in several phases (initiation, problem definition, role assignment, idea generation, criteria development, decision, implementation, presentation, written evaluation), the defined objective of a project assignment within a given period of time and using suitable instruments. ²In addition, project work may include a presentation in order to assess a student's communication competency in presenting scholarly work to an audience. ³Project work may also include creative drafts, drawings, plans, models, objects, simulations and documentation.
- e) ¹A **research paper** is a written assignment in which students work independently on solving complex scholarly or scholarly/application-oriented problems, using the scientific methods of the related discipline. ²Students are expected to demonstrate that they are able to solve problems corresponding to the learning results of the module in question in compliance with the guidelines for scholarly work – from analysis and conception to implementation. ³Research papers, differing in their requirement standards, may take the form of a conceptual framework/theory paper [Thesenpapier], abstract, term paper, seminar paper, etc. ⁴The research paper may be complemented by a presentation and/or a colloquium for the purpose of assessing the student's communication competency in presenting scholarly work to an audience.
- f) ¹A **presentation** is a systematic and structured oral performance supported by suitable audio-visual equipment (such as beamer, slides, posters, videos) for the purpose of demonstrating and summarizing specific issues or results and paring complex problems down to their essential core. ²For the presentation, the student is expected to demonstrate that he or she is capable of preparing a certain topic within a given time frame in such a way as to present or report it in a clear and comprehensible manner to an audience. ³In addition, the student is expected to demonstrate that he or she is able to respond competently to any questions, suggestions or discussions brought by the audience and relating to his or her subject area. ⁴ The presentation can be supplemented by a short written presentation.
- g) ¹An **oral examination** is a timed, graded discussion on relevant topics and specific questions to be answered. ²The purpose of oral examinations students are expected to demonstrate that they have reached the qualification objectives laid out in the module descriptions, understood the central concepts of the subject matters covered by the exam, and are able to apply them to specific problems. ³The duration of the examination is provided for in § 13 (2) of the APSO.
- h) ¹A **learning portfolio** is a collection of written materials compiled by the student according to predefined criteria that exhibits the student's progress and achievements in defined content areas at a given time. ²Students are required to explain according to which criteria they have chosen the materials and their relevance for their learning progress and the achievement of the qualification objectives. ³With the learning portfolio, students are expected to demonstrate that they have taken active responsibility for their learning process and have reached the qualification objectives set out in the module description. ⁴Depending on the module description, types of independent study assessment in a learning portfolio

may include, in particular, application-oriented assignments, web pages, weblogs, bibliographies, analyses, conceptual framework/theory papers, as well as the graphic representation of facts or problems. ⁵The specific components of each learning portfolio and the related competencies to be assessed are set out in the module description.

- (2) ¹The module examinations will, as a rule, be taken concurrently with the program. ²Type and duration of module examinations are provided for in Appendix 1 (II). ³In the event of divergence from those provisions, § 12 (8) of the APSO must be complied with. ⁴The assessment of the module examination is governed by § 17 of the APSO. ⁵The weights of the individual examinations are determined in appendix 1 (II) according to the credits.
- (3) Where Appendix 1 (II) provides that a module examination is either in written or oral form, the examiner will inform the students in appropriate form, at the commencement of classes at the latest, of the type of examination to be held.

§ 42

Registration for and Admission to the Master's Examination

- (1) Students who are enrolled in the Master's program in Management (Campus Heilbronn) are deemed admitted to the module examinations of the Master's examination.
- (2) ¹Registration for a module examination is governed by Section 15 (1) APSO. ²Registration for a corresponding repeat examination is governed by § 15 Para. 2 APSO.

§ 43

Scope of the Master's examination

- (1) The Master's examination consists of:
 - 1. the module examinations in the corresponding modules pursuant to subsection (2)
 - 2. the Master's thesis pursuant to § 46.
- (2) ¹The module examinations are listed in Annex 1 (II.). ²Students must earn 60 credits in the compulsory modules and at least 30 credits in elective modules in the area of Electives in Management, of which at least 6 credits must be earned through an Advanced Seminar. ³When selecting the modules, § 8 Para. 2 APSO must be observed.

§ 44

Repeat Examinations, Failed Examinations

- (1) The repetition of examinations is governed by § 24 of the APSO.
- (2) Failure of examinations is governed by § 23 of the APSO.

§ 45

Coursework

The Master's program in Management at the Campus Heilbronn does not require coursework, only the successful completion of examinations.

§ 45 a Multiple-Choice Test

The undertaking of multiple-choice tests is governed by § 12 of the APSO.

§ 46 Master's Thesis

- (1) ¹In accordance with § 18 APSO, students must complete a thesis as part of the Master's examination in the Master's Thesis module.
- (2) ¹As a rule, completion of the Master's Thesis module should be the final examination ²Students may be admitted to the Master's Thesis module early upon application if at least 45 out of 60 credits from the compulsory modules and 6 credits from the elective modules have already been earned and if the objective of the thesis can be achieved within the meaning of Section 18 (2) APSO, taking into account the course of study to date.
- (3) ¹The period of time between topic assignment and submission of the completed Master's thesis must not exceed six months. ²The thesis is considered presented and not passed if the student fails to submit it on time without valid reasons pursuant § 10 (7) of the APSO. ³The module is worth 30 credits. ⁴The Master's thesis should be written in English language.
- (4) The completion of the Master's Thesis module consists of a scientific paper.
- (5) ¹If the Master's thesis was not graded with at least „sufficient“ (4.0), it may be repeated once with a new topic. ²Students must renew their application for admission within six weeks from receipt of the grade.

§ 47 Passing and Assessment of the Master's Examination

- (1) The Master's Examination is deemed passed when all examinations required for the Master's examination pursuant to § 43 (1) have been passed and a plus credits account of at least 120 credits has been achieved.
- (2) ¹The grade of a module will be calculated according to § 17 of the APSO. ²The overall grade for the Master's examination will be calculated as the weighted grade average of the modules according § 43 (2) sentence 2 and the Master's thesis. ³The grade weights of the individual modules correspond to the credits assigned to each module. ⁴The overall assessment is expressed by the designation pursuant §17 of the APSO.

§ 48 Degree Certificate, Diploma, Diploma Supplement

¹If the Master's examination was passed, a degree certificate, a diploma, and a diploma supplement including a transcript of records are to be issued in compliance with § 25 (1) and § 26 of the APSO.

§ 49*) Entry into Force

¹These statutes come into force on January 1, 2022. ²It applies to all students who begin their studies at the Technical University of Munich from the winter semester 2022/2023 onwards.

¹The subject examination and study regulations for the Master's degree program in Management at the Heilbronn campus of the Technical University of Munich dated June 27, 2018, in the version of No. 75 of the collective amendment statutes on the number of examining committee members in the aptitude procedure for Master's degree programs at the Technical University of Munich dated June 29, 2020, shall expire at the same time. ²Students who have already started their studies at the Technical University before the winter semester 2022/2023 will complete their studies in accordance with the statutes pursuant to sentence 1.

*) This provision relates to the entry into force of the Articles of Association in the original version dated November 15, 2021. The date of entry into force of the subsequent amendments is determined by the respective amending Articles of Association.

Appendix 1:

I. Scope of the Master's Examination

	Components	Credits	Semester
1.	Study accompanying examinations for the achievement of the required credits in basics in management	42	1./2. Semester
2.	Study accompanying examinations for the achievement of the required credits in basics in economics	6	1. Semester
3	Study accompanying examinations for the achievement of the required credits in basics in law	6	2. Semester
4.	Study accompanying examinations for the achievement of the required credits in in Entrepreneurial, Strategic, and International Management	6	1. Semester
5.	Study accompanying examinations for the achievement of the required credits in electives in management	30	3. Semester
6.	Master's Thesis pursuant to § 46	30	4. Semester
		120	

II. Exam modules

Master in Management (Required modules)

The following required modules have to be passed successfully.

module number	module name	Required / required elective	Type / weekly hours V Ü P S	Sem.	SWS	Credits	Type of assessment ²⁾	Duration of exams	Language of instruction
Basics in management									
WIHN1139	Financial Accounting (MiM)	Required	2 V + 2 Ü	1. Sem.	4	6 Credits	Klausur	60	English
WIHN1137	Management Science (MiM)	Required	2 V + 2 Ü	1. Sem.	4	6 Credits	Klausur	60	English
WIHN0258	Empirical Research in Economics and Management	Required	2 V + 2 Ü	1. Sem.	4	6 Credits	Klausur	120	English
WIHN1138	Investment and Financial Management (MiM)	Required	2 V + 2 Ü	2. Sem.	4	6 Credits	Klausur	120	English
WIHN1129	Marketing & Innovation Management (MiM)	Required	4 V	2. Sem.	4	6 Credits	Klausur	120	English
WIHN1131	Production and Logistics (MiM)	Required	4 V	2. Sem.	4	6 Credits	Klausur	120	English
WIHN1130	Cost Accounting (MiM)	Required	2 V + 2 Ü	2. Sem.	4	6 Credits	Klausur	60	English

Basics in Economics									
WIHN1056_1	Principles of Economics	Required	2 V + 2 Ü	1. Sem.	4	6 Credits	Klausur	120	English

Basics in Law									
WIHN1122	Introduction to Business Law (MiM)	Required	2 V + 2 Ü	2. Sem.	4	6 Credits	Klausur	120	English

Entrepreneurial, Strategic and International Management									
MGTHN0102	Entrepreneurship	Required	2 V	1. Sem.	2	3 Credits	Klausur	60	English
MGTHN0103	Strategic and International Management	Required	2 V	1. Sem.	2	3 Credits	Klausur	60	English

Electives in management

As part of the economics electives, students complete credits in economics from a supplementary elective catalog totaling at least 30 credits. Of these, 12 credits can be earned as part of a project study in accordance with § 37 a.

Instead of examinations at TUM, subject-related examinations amounting to 30 credits can be taken at a foreign university as part of a stay abroad. For this purpose, the students compile an individual

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semester study plan with a mentor appointed by the faculty. The corresponding courses are to be selected from the courses offered by the foreign university.

The supplementary elective catalog will be announced in an appropriate manner by the Faculty of Business and Economics on time before the start of lectures; enclosed is an exemplary catalog of electives.

module-number	module name	required / elective	type weekly hours V Ü P S	Sem.	SWS	Credits	Type of assessment ²⁾	duration of exam	language of instruction
WIHN9684	Advanced Project Studies ¹⁾	Elective		3./4. Sem.	8	12 Credits	project	n.a.	English
WIHN0017	Advanced Seminar in Innovation and Entrepreneurship Digital Innovation	Elective	4 S	3./4. Sem.	4	6 Credits	report	n.a.	English
WIHN0045	Advanced Seminar Innovation & Entrepreneurship: Strategic Decision-Making in Entrepreneurship and Family Enterprises	Elective	4 S	3./4. Sem.	4	6 Credits	Report	n.a.	English
WIHN0011	Advanced Seminar Operations & Technology	Elective	4 V	3./4. Sem.	4	6 Credits	report	n.a.	English
WIHN0012	Digital Finance	Elective	4 VI	3./4. Sem.	4	6 Credits	Klausur	90	English
MGTHN0075	Cross-Cultural Management	Elective	2V + 2Ü	3./4. Sem.	4	6 Credits	Klausur	120	English
WIHN0014	Corporate Campus Challenge	Elective	4 S	3. Sem.	4	6 Credits	report	n.a.	English
WIHN0013	Advanced Seminar Finance & Accounting: Current Research Topics in Digital Finance	Elective	4 S	3./4. Sem.	4	6 Credits	report	n.a.	English

Annotation:

- 1) This module with the associated partial module examinations can extend over at least two semesters.

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Master's Thesis

module number	Master's Thesis	Required / elective				Credits			Language of instruction
WIHN9268	Master's Thesis (Master in Management)	Required		4. Sem.		30 Credits	report	n.a.	English

Explanations:

Sem. = Semester; SWS = Semesterwochenstunden (weekly hours); V = Vorlesung (Exam); Ü = Übung (exercise); K = Klausur (exam); S = Seminar. In case of an oral or written exam, the duration of the exam can be found in the column "duration of exam" (in minutes).

III. Study plan

1. Study plan

Semester	Event	module type	credits
1. Semester			
	Financial Accounting (MiM)	required module	6
	Management Science (MiM)	required module	6
	Empirical Research in Economics and Management	required module	6
	Principles of Economics	required module	6
	Entrepreneurship	required module	3
	Strategic and International Management	required module	3
Credits totaling 1. semester:			30
2. Semester			
	Investment and Financial Management (MiM)	required module	6
	Introduction to Business Law (MiM)	required module	6
	Marketing & Innovation Management (MiM)	required module	6
	Production and Logistics (MiM)	required module	6
	Cost Accounting (MiM)	required module	6
Credits totaling 2. semester:			30
3. Semester			
	Electives in management	elective	30
Credits totaling 3. semester:			30
4. Semester	Master's Thesis (Master in Management)	required module	30
Credits totaling 4. semester:			30
Total Credits Master's in Management (Campus Heilbronn):			120

Appendix 2: Aptitude Assessment

Aptitude Assessment for the Master's Program in Management at the Heilbronn campus of Technische Universität München

1. Purpose of the Assessment

¹Qualification for the Master's degree program in Management at the Heilbronn Campus at the Technical University of Munich requires, in addition to the requirements pursuant to § 36 (1) no(s) 1 to 3, proof of aptitude in accordance with § 36 (1) no. 4 in accordance with the following provisions.²The special qualifications and skills of the candidates should correspond to the profession of engineering or natural sciences. ³Individual aptitude parameters are:

- 1.1 Subject knowledge (incl. success) from a first degree in the field of engineering or natural sciences in accordance with the engineering and natural science bachelor's degree programs of the Technische Universität München.
- 1.2 Knowledge in the economic-technical context
- 1.3 Ability to do research work and/or basic research and methodological work;
- 1.4 English technical language proficiency

2. Aptitude Assessment Process

2.1 ¹The aptitude assessment procedure takes place every six months. ²The Aptitude Assessment Procedure is governed by the Statutes of the Technical University of Munich on Enrollment, Re-registration and De-registration (ImmatS) of February 6, 2023, as amended, in particular § 6.

2.2 ¹Applications for the aptitude assessment procedure in accordance with § 6 ImmatS must be submitted to the Technical University of Munich together with the documents listed, the documents listed under no. 2.3 and the documents in accordance with § 36 para. 1 no. 2 and 3 for the winter semester using the online application procedure by May 31 (cut-off deadline). ²The certificate and the transcript as proof of passing the Bachelor's degree program must be submitted to the TUM Center for Study and Teaching - Application and Enrollment no later than five weeks after the start of lectures. ³Failing this, admission to the Master's degree program is not yet possible in accordance with § 36 of these statutes.

2.3 The application must include:

- 2.3.1 transcript of records with modules totaling 140 credits, issued by the respective examination board or the responsible office of student affairs and handed in as certified copy.
- 2.3.2 a curricular analysis derived from the transcript of records; this must be completed by the applicants as part of the online application process and uploaded electronically on TUMonline.
- 2.3.3 curriculum vitae in resume form
- 2.3.4 a written essay in English of at least 1500 and maximum of 2000 words; the essay must be uploaded as a PDF document to TUMonline; the chairperson of the committee may provide one topic or a selection of several topics for this essay. The candidates must be informed of the topic(s) not later than 1 April.
- 2.3.5 an assurance that the essay was written independently and without outside assistance and in compliance with the guidelines for safeguarding good scientific practice and for dealing with scientific misconduct at the Technical University of Munich and that the ideas taken from outside sources are marked as such.

- 2.3.6 optionally, for applicants who are not required to provide proof of a GMAT score in accordance with Section 36 (1) No. 3, proof of a GMAT score.

3. Aptitude Assessment Committee

- 3.1 ¹The aptitude assessment procedure is carried out by the aptitude assessment committee and the selection committees. ²The committee for the aptitude assessment procedure is responsible for preparing the procedure, organizing it and ensuring a structured and standardized procedure for determining aptitude within the framework of these statutes; it is responsible insofar as no other responsibility is stipulated by these statutes or by delegation. ³The selection committees are responsible for carrying out the procedure in accordance with no. 5, subject to no. 3.2 sentence 11.
- 3.2 ¹The committee for the aptitude assessment procedure (committee) consists of five members. ²These are appointed by the Dean in consultation with the Vice Dean Academic and Student Affairs from among the members of the TUM School of Management who are eligible to take examinations and are involved in the degree program. ³At least three of the committee members must be university lecturers within the meaning of the BayHIG. ⁴The student council has the right to nominate a student representative to serve on the committee in an advisory capacity. ⁵A deputy shall be appointed for each member of the commission. ⁶The committee elects a chairperson and a deputy chairperson from among its members. ⁷The procedural provisions of the TUM's Basic Regulations, as amended, shall apply to the course of business. ⁸The term of office of the members is one year. ⁹Reappointment and extension of the term of office are permitted. ¹⁰The Chairperson can make urgent decisions that cannot be postponed on behalf of the aptitude assessment committee; he or she must inform the committee immediately. ¹¹The Office of Student Affairs supports the Commission for the Aptitude Assessment Procedure and the selection committees; the Commission for the Aptitude Assessment Procedure can assign the Office of Student Affairs the task of the formal admissions evaluation in accordance with section 4, as well as the evaluation of points on the basis of previously defined criteria where there is no scope for evaluation, in particular the conversion of grades, the determination of the total number of points obtained, the composition of the Selection Committees from the members appointed by the Admissions Committee, and the allocation of points to the applicants.
- 3.3 ¹The selection committees each consist of two members from among the members of the TUM School of Management who are eligible to take examinations in the degree program in accordance with Art. 85 Para. 1 Sentence 1 BayHIG in conjunction with the University Examination Regulations. ²At least one member must be a university lecturer within the meaning of the BayHIG. ³The activity as a member of the Admissions committee can be carried out alongside the activity as a member of the selection committee. ⁴The members are appointed by the aptitude test committee for one year; no. 3.2 sentence 9 applies accordingly. ⁵Different selection committees may be appointed for each criterion and level.

4. Admission to the Aptitude Assessment Process

- 4.1 ¹The eligibility procedure requires that the documents specified in no. 2.2 are submitted in due form and time, in compliance with the rules of good scientific practice and in full. ²In order to determine whether the rules of good scientific practice have been observed, the essay will be checked using special plagiarism checking software.
- 4.2 ¹Applications who fulfills the necessary requirements according to No. 4.1 will be examined in the aptitude procedure according to No. 5. ²Otherwise, a rejection notice will be issued, stating the reasons for the rejection and informing the applicant of his or her rights of appeal.

³If the selection committee comes to the conclusion that the rules of good scientific practice have been seriously violated, the applicant will be excluded from the current application procedure. ⁴Sentence 2 applies accordingly.

5. The Aptitude Assessment Process

5.1 First stage of the aptitude assessment process.

- 5.1.1 ¹The committee will assess, on the basis of the written application documents required under no. 2.3, whether or not an applicant is suitable for a program pursuant to no. 1 (first stage of the aptitude assessment process). ²For this purpose, the committee evaluates and grades the candidate's documentation on a scale ranging from 0 to 75, 0 being the worst and 75 being the best possible result. ³Negative results will not be given.

⁴The following criteria will be applied to the evaluation:

a) Academic qualification

¹The curricular analysis of the existing specialized knowledge is not carried out by schematic comparison of the modules, but on the basis of competencies. ²It is based on the elementary subject groups listed below, which are taken into account either for an engineering bachelor's degree program or for a bachelor's degree program in the natural sciences.

Subject groups Bachelor's program in engineering

a) Engineering Fundamentals:

Mathematics, Technical mechanics, Machine Design, Materials Science, Machinery and Plant Engineering,

b) Bachelor Electrical Engineering and Information Technology:

Mathematics, computer science, digital technology, circuit technology, electricity and magnetism, physics, measurement system and sensor technology, signals, materials of electrical engineering, electromagnetic field theory, systems, electronic components, computer technology, electrical power engineering.

c) Bachelor of Civil Engineering and Surveying:

Mathematics, engineering mechanics, hydromechanics, construction process management, structural analysis, finite elements, hydrology, hydraulic engineering, urban water management, surveying.

d) Bachelor architecture

Design, construction, statics and strength of materials, history of construction, representation and design, structural design and materials, building climates and building technology, digital form finding, urban design, urban planning, urban studies, visual design, architecture and design theory, art history.

e) Other engineering subject groups

For engineering bachelor degrees other than those listed under a) to d), the subject catalog of the corresponding degree program at the Technical University of Munich is used as a basis.

Subject groups in natural sciences

a) Bachelor Computer Science:

Introduction to Computer Science, Computer Engineering, Discrete Structures, Fundamentals of Programming, Algorithms and Data Structures, Databases, Operating Systems and Systems Software, Analysis, Computer Networks and Distributed Systems, Theoretical Computer Science, Discrete Probability Theory, Numerical Programming, Systems Development;

b) Bachelor mathematics:

Analysis, Linear Algebra, Discrete Mathematics, Numeric, Probability Theory, Mathematical Modeling, Programming Language;

c) Bachelor Games Engineering:

Introduction to Computer Science for Games Engineering, Fundamentals of Programming, Introduction to Computer Science, Discrete Structures, Games Engine Design, Introduction to Software Engineering, Algorithms and Data Structures, Linear Algebra, Interaction Methods and Devices, Aspects of System-Oriented Programming in Game Development, Operating Systems and Hardware-Oriented Programming for Games, Analysis, Social Gaming, Computer Networks and Distributed Systems, Theoretical Computer Science, Numerical Programming, Physical Principles for Computer Games, Databases, Fundamentals of Artificial Intelligence;

d) Bachelor Bioinformatics:

Introduction to Bioinformatics, Introduction to Programming, Introduction to Computer Science, Fundamentals of Programming, Analysis, Discrete Structures, Biology, Chemistry, Programming and Modeling, Algorithms and Data Structures, Discrete Structures, Linear Algebra, Analysis, Biochemistry, Molecular Biology and Biochemistry, Algorithmic Bioinformatics, Formal Languages and Complexity, Theoretical Computer Science, Stochastics, Discrete Probability Theory, Genome-Oriented Bioinformatics, Databases, Advanced Bioinformatics;

e) Bachelor Biology:

Mathematics, Inorganic Chemistry, Biology of Organisms, Zoology, Cell Biology, Genetics, Organic Chemistry, Physical Chemistry, Computer Science, Statistics, Experimental Physics, Microbiology, Bioinformatics, Biochemistry, Ecology, Human and Animal Physiology, Botany, Plant Physiology, Bioanalysis, Evolution and Biodiversity, Developmental Biology, Genomics and Genetic Engineering;

f) Bachelor Physics:

Experimental physics, linear algebra, calculus, theoretical physics (mechanics/electrodynamics/quantum mechanics), and chemistry;

g) Bachelor Chemistry

Inorganic Experimental Chemistry, Basic Inorganic Chemical Practicum, Experimental Physics, Principles and Methods of Chemistry, Mathematical Methods of Chemistry, Biology, Analytical Chemistry, Structure of Organic Compounds, Chemical Thermodynamics and Kinetics, Chemistry of Nonmetals, Preparative Inorganic Chemical Practicum, Reactivity of Organic Compounds, Biochemistry, Physical Chemistry Practicum on Thermodynamics, Introduction to Quantum Mechanics, Structural Analytical Techniques, Chemistry of Metals and Inorganic Solids, Organic Chemistry Practicum, Molecular Structure and Statistical Thermodynamics, Trace Analytical Techniques, Toxicology;

h) Bachelor Nutritional Science:

Inorganic experimental chemistry, Inorganic chemistry, Cell biology, Experimental physics, Physical practical, Mathematics, Biology: Genetics, Human Physiology, Computer Science, Organic Chemistry, Principles of Human Nutrition, Food Science, Biochemistry, Physiology, Morphology, Microbiology, Nutritional Physiology of Macro- and Micronutrients, Food Microbiology, Food Science, Immunology, Experimental Nutrition Research, Nutritional Medicine, Human Sensory Science, Bio functionality of Food, Toxicology, Pharmacology and Clinical Studies, Regulation of Metabolism, Biostatistics;

i) Bachelor Earth sciences:

Mathematics, Experimental Physics, Chemistry, Biology, Data Processing in the Geosciences, General Mineralogy, Paleontology, Geological Maps and Profiles, Rocks, Applied Geophysics;

j) Bachelor Biotechnology:

Inorganic Experimental Chemistry, Biochemistry, Software and Databases, Basic Biochemistry Lab, Basic Chemistry Lab, Introduction to Genetics, Mathematics, Physics, Plant Science, Computer Science, Microbiology, Organic Chemistry, Physiology and Functional Anatomy, Protein Biochemistry, Bioinformatics/Genomics/Proteomics, Bio Inorganic Chemistry, Statistics, Physical Chemistry, Biochemical Analysis, Biotechnology, Immunology, Cellular Biochemistry, Bioprocess Engineering, Plant Molecular Biology, Molecular Bacterial Genetics, Molecular Plant Breeding, Molecular Genetics and Regulatory Physiology of Animals, Proteins: Structure, Function and Engineering, Metabolic Engineering and Natural Product Production, Pharmacology and Toxicology;

k) Bachelor Scientific basics of sport:

Basic competencies in sports science (introduction to sports science work, basics of the theory of sports science, basics of mathematics and natural sciences), anatomical basics for sports and health science (functional anatomy of the human musculoskeletal system, anatomy of the internal organs), biological/physiological basics (biochemistry, physiology), anatomical basics for sports and health science, movement sciences (basics of movement science, basics of biomechanics), methodology (experimental design and descriptive statistics, Probability theory and inferential statistics, Experimental work), Biological/physiological competencies in sport (neuroanatomy/physiology, performance/work physiology, nutrition and fluid balance, doping prevention), Health science, Fundamentals of sports medicine, Introduction to traumatology, Training Science Competency, Psychological Competency (Fundamentals of Psych regulation, Psychology of Motivation and Emotion), Sports Education/Didactic Competency, Applied Anatomy, Biomechanical Competency, Basic Sports Diagnostic Competency, Sports Science Methodological Competency, Sports Technology;

l) Bachelor Medicine:

General medicine, anesthesiology, occupational medicine, social medicine, ophthalmology, surgery, dermatology, venereology, gynecology, obstetrics, otolaryngology, human genetics, hygiene, microbiology, virology, internal medicine, pediatrics, clinical chemistry, laboratory diagnostics, neurology, orthopedics, pathology, pharmacology, toxicology, psychiatry and psychotherapy, psychosomatic medicine and psychotherapy, legal medicine, urology.

³Where a candidate's competencies are at least equivalent to those listed above he/she will be awarded a maximum of 50 points. ⁴In the case this value is not an integer, it shall be rounded up to the next larger number. ⁵Missing competencies will be deducted in accordance with the credits of the corresponding modules of the respective Bachelor's program at the Technische Universität München. ⁶If a GMAT score is to be submitted in accordance with § 36 Para. 1 No. 3, it will be assumed that there are no significant differences in the level of the competencies demonstrated in the first degree compared to the reference degree programs mentioned in Section 5.1.1 a) Sentence 2 and that the curricular analysis is carried out in accordance with the above criteria.

b) Final Grade

¹One point is awarded for each tenth grade that is better than 2.5 in the average calculated over 140 credits. ²The maximum number of points is 15. ³In the case of foreign degrees or if the grading system does not correspond to that of TUM, the grade converted using the Bavarian formula will be used. ⁴If a degree certificate with more than 140 credits is available at the time of application, the assessment will be based on the best graded modules amounting to 140 credits. ⁵It is the responsibility of the applicant to list these in the application and to confirm in writing that the information is correct. ⁶If this is done, the average grade will be calculated from the best graded module examinations amounting to 140 credits; if this information is missing, the overall average grade submitted by the applicant will be used. ⁷The overall grade point average is calculated as the weighted grade average of the modules. ⁸The grade weights of the individual modules correspond to the assigned credits.

Note	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2	2.1	2.2	2.3	2.4	≥ 2.5
Points	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

c) GMAT-Score

¹The proof of a current GMAT-Score with at least 650 points will be awarded with 1 point. ²All other scores are evaluated as follows:

GMAT	≤640	650	660	670	680	690	700	≥710
Points	0	1	2	3	4	6	8	10

5.1.2 ¹The applicant's points total are calculated as sum of the individual points awarded. ²The decimal places that are not being scratched must be rounded up.

5.1.3 ¹Applicants who achieved at least 62 points will receive confirmation that they have passed the aptitude assessment test.

5.1.4 ¹Unsuitable applicants with an overall grade of fewer than 50 points have failed the aptitude test.

5.2 Second stage of the aptitude test

5.2.1 ¹The remaining applicants will be evaluated according to their essay. ²In the second stage of the aptitude assessment process the applicant's qualification at undergraduate level, the grade of the ranking (or the overall grade) and the result of the essay will be evaluated, taking at least equal consideration of the qualification obtained at undergraduate level. ³The essay will be graded with points between 0 to 40 by two members of the committee. ⁴The content of the essay will be graded according to the following criteria:

The applicants should be able to:

1. to present skills of scientific-logical argumentation with basic and method-oriented text structure, as well as to write the essay in a scientific manner and with correct citation of sources,
2. to frame the question in the context of economic-technical issues,
3. to express themselves in English.

5.2.2 All three criteria will be graded individually by each member of the committee. The criteria will be weighted as follows:

1. Ability to present scientific logical argumentation with basic and methodological text structure as well as to write the essay in a scientific manner and with correct citation of sources: maximum 20 points,
2. Framing of the question in the context of economic-technical issues: maximum 10 points,
3. Expression in English: 10 Punkte.

5.2.3 ¹The essay is evaluated by the selection committee. ²The two members of the selection committee evaluate each of the three key areas independently. ³The score per committee member results from the sum of the weighted evaluations of the individual criteria. ⁴The total score is the arithmetic mean of the scores of the two committee members, rounded up to whole numbers of points. ⁵The maximum number of points is 40.

5.2.4 ¹The applicant's points total in the second stage are calculated as the sum of all points obtained under 5.2.3 and the points under 5.1.1.a) (academic qualification) and 5.1.1. b) (final grade). ²Applicants with 80 or more points will be deemed suitable. ³Candidates with a total application score of less than 80 points have failed the aptitude assessment procedure.

5.2.5 ¹The result of the aptitude test is determined on the basis of the number of points achieved and announced in a decision. ²A rejection notice must specify the reason for the rejection and provide information on legal remedies.

5.2.6 Admissions to the Master's program in Management shall apply to all subsequent applications for this program.

6. Documentation

The aptitude assessment procedure must be documented; in particular, the documentation must include the names of the persons involved in the decision, the names of the applicants, the assessment of the first and second stages, the overall result and the main reasons for the assessment of the essay; the main reasons may be listed in keywords.

7. Repetition

Applicants who have failed the aptitude test for the Master's program in management may register for one repetition of the Aptitude Assessment Test.